ShikkhaSetu: A Connectivity-Tiered Blended Learning System That Keeps Bangladeshi Children Learning When Schools Close

1. Title and Brief Summary

Bangladesh has expanded school access, but learning remains weak and schooling is repeatedly interrupted. Schools were closed for 543 days during COVID-19, and in 2024 heatwaves, Cyclone Remal and floods disrupted schooling for 35 million children [8, 9]. When schools close, the remote options mostly assume a television, internet or smartphone, and large surveys show that only a small share of students used them [1, 13]. **ShikkhaSetu** (Bangla for "learning bridge") is a blended learning concept in which a school's own teacher sends each weekly lesson through **three channels at once**: an offline-first smartphone app, a basic-phone voice lesson with SMS quiz, and a printed worksheet. On normal days it supports classroom learning with home practice. When a disaster closes schools, a single **Remote Mode** switch keeps every learner connected to the same teacher. The concept is untested; this note presents the evidence, the gap, the concept design and a feasibility plan for a small pilot.

2. Problem Statement and Target Users

Problem: Many Bangladeshi children are not achieving basic learning, their schooling is frequently interrupted, and the remote learning that exists does not reach most of the children who need it. Verified evidence:

- **Weak foundational learning:** 51% of children could not read and understand a simple text by age 10 in 2022 (World Bank learning poverty measure) [14].
- **Repeated disruption:** 543 consecutive days of closure (17 March 2020 to 12 September 2021) affected more than 35 million students [8]. In 2024, nationwide heatwave closures of up to two weeks, Cyclone Remal and June floods disrupted schooling for 35 million children; children in Sylhet lost up to 8 weeks of school days [9].
- **Low reach of remote learning:** a BIDS survey of 50,871 students in 63 districts found only 9 to 11% of primary and 16.2% of secondary students engaged with televised lessons during closures [1]. Among poor Grade 9 stipend recipients, only 21% had access to government online programmes and about 2% of those used them; 1 in 2 parents said they could not help with new topics [13].
- **Device gap:** 99% of households have a mobile phone and 74.9% have a smartphone, but only 59.2% have internet access (56.4% in rural areas) and only 3.7% of rural households have a computer (BBS ICT Access and Use Survey 2024-25) [4, 5].

Target users: students in Grades 4 to 8 in rural and climate-vulnerable upazilas, especially from poorer households; their teachers, who lose contact with learners during closures; parents (often mothers, who were found to be more involved than fathers [13]) who are asked to support learning without guidance; and head teachers and upazila education officers who need to know which children are falling behind. **Why it matters:** SDG 4 requires inclusive and equitable quality education. Each closure deepens an existing learning crisis, and the children least able to learn remotely are those from the poorest and most rural households [8, 11].

3. Existing Solutions and Research Findings

Solution	What it does	What it does well	What could be improved for this problem
Government TV and online classes (Sangsad TV, Konnect)	National broadcast of recorded lessons during closures.	Free, national scale, launched within weeks of closure [13].	One-way and needs a TV; only 9 to 16% engagement [1]; no contact with the child's own teacher.
10 Minute School (private EdTech)	Video lessons, live classes and quizzes for the national curriculum.	One of the two dominant K-12 EdTech players [3]; reports 17M+ students (self-reported) [16].	Needs a smartphone and data; premium content is paid; not linked to the child's school or teacher.
Muktopaath and Teachers' Portal (a2i, government)	Bangla e-learning for teacher training, and a content-sharing network of 600,000+ registered educators [10].	Proven teacher reach: 96.47% of enrolled teachers completed the national curriculum course [2].	Built for teacher training, not daily teacher-to-learner contact; classroom use of training is uneven [12].
Kolibri (Learning Equality, global)	Free, open-source, offline-first platform with a local school server and sync.	Works without internet; teacher dashboards; installed in 220+ countries and territories [7].	Needs tablets or computers; does not reach homes that only have a basic phone.
Phone call + SMS tutoring (Botswana trial, global)	Weekly SMS problems plus a 15 to 20 minute call to about 4,500 families.	+0.12 SD learning, 31% less innumeracy; among the most cost-effective options (up to 0.89 SD per US$100) [6].	SMS alone had no significant effect; tested on numeracy only; not integrated with schools.

Research caution: a Bangladesh trial of phone mentoring reported large gains [15], but a 2025 Institute for Replication comment raised serious data-integrity concerns [17], so this note does not rely on it. **Key findings:** (1) channels that need TV or internet reached a minority of students; (2) live human contact by phone produced gains where one-way messages did not; (3) Bangladesh already has the building blocks: curriculum content, national teacher platforms, near-universal mobile phones and a National Blended Education Master Plan (2022-2041) that explicitly combines high-, low- and no-tech learning [11].

4. Identified Gap or Opportunity

Verified findings: remote learning reached a minority of Bangladeshi students [1, 13]; mobile phones are near-universal while home internet and computers are not [4, 5]; and phone plus SMS support has produced cost-effective gains in a large trial [6]. **Gap (an inference from this research, not yet validated with users):** none of the solutions reviewed combines (a) the child's own school teacher, (b) the same lesson delivered across smartphone, basic phone and paper, and (c) a ready switch between in-school blended mode and full remote mode when disasters strike. Each existing tool covers one channel or one user group.

Opportunity in Bangladesh: a low-cost layer that plugs into existing government platforms and the Blended Education Master Plan, aimed at the upazilas that lose the most school days to floods and heat [9, 11]. **Opportunity worldwide:** at least 463 million schoolchildren (31%), including 147 million in South Asia, could not be reached by remote learning during COVID-19 [18], and in 2024 at least 242 million students in 85 countries had schooling disrupted by climate events, 74% of them in low- and lower-middle-income countries [19]. A phone-first continuity model could transfer to similar settings. **Assumptions still to test:** teachers can spare about one extra hour per week; families will answer calls and share a guardian number; and SMS and voice costs stay affordable at scale.

5. Proposed Solution and Key Features

How it works: each week the teacher selects a curriculum-aligned lesson from a shared library, and ShikkhaSetu packages it for every channel. Learners study at home on whatever device the family has, answer a short quiz, and the results flow back to a single class dashboard. In-school days are used for practice, discussion and support for the learners the dashboard flags.

- **One Lesson, Three Channels:** an offline app pack (short video and quiz, downloaded once over school Wi-Fi), a 5 to 7 minute Bangla voice lesson by phone with a 3-question SMS quiz, and a printable worksheet.
- **Learning Level Check:** a short phone or app check every two weeks groups learners into three levels so teachers can target support instead of teaching only to the grade.
- **Weekly Mentor Call:** a 15 minute call to learners flagged red (no activity or low scores) by the teacher or a trained volunteer, following the phone plus SMS evidence [6].
- **Remote Mode switch:** one tap by the head teacher sends a voice and SMS alert to parents, publishes the home-learning schedule and increases call frequency. On reopening it runs a catch-up check.
- **Parent Nudges and Progress Dashboard:** a weekly voice message suits parents with low literacy; the dashboard shows green, yellow or red status per learner across all channels and rolls up to upazila level.

Value: students keep learning on any phone or on paper; teachers stay in contact; parents get simple guidance; and education officers see who is falling behind before it becomes dropout.

6. Visual Representation

Figure 1 shows the design context. Learning normally happens in the classroom (a). When a flood or heatwave closes schools (b), the mobile network often still reaches even remote wetland (haor) villages (c). ShikkhaSetu uses that network, plus paper, to bridge (a) and (b), so the same teacher and weekly lesson continue in both states.



(a) School open: rural classroom



(b) School closed: 2024 flood, Chattogram



(c) Network still reaches: haor area

Figure 1. Design context for ShikkhaSetu. Photo panel adapted by the author from openly licensed Wikimedia Commons photos: (a) Mark Knobil, CC BY 2.0; (b) Owais Al Qarni, CC BY-SA 4.0; (c) Balaram Mahalder, CC BY-SA 3.0 [20 to 22].

7. Feasibility, Risks and Next Steps

Area	Initial assessment for a 6-month pilot (10 schools, one flood-prone upazila)
Resources and skills	Product designer, Android developer, backend developer, curriculum specialist (experienced teacher), field coordinator, part-time monitoring and evaluation researcher; volunteer mentors (for example university students).
Tools	Offline-first Android app; open-source Kolibri evaluated as the offline content and sync layer [7]; SMS and voice (IVR) gateway through a mobile operator or aggregator; a low-cost school laptop or hotspot as sync hub; school printer.
Time	Month 1: co-design with 5 teachers and 20 families. Months 2 to 3: build MVP and train teachers. Months 4 to 5: run the pilot, including one simulated Remote Mode week. Month 6: evaluate learning, usage and cost per learner.
Dependencies	Permission from the Directorate of Primary Education or DSHE and the upazila education office; content licensing; an operator agreement for a toll-free or zero-rated short code (requires regulatory approval); an NGO or a2i partner; seed funding.
Benefits	Reaches basic-phone households; keeps the teacher-learner link; prepares schools for disasters; gives data to target support.

Key risks and mitigations:

- **Low call pickup or a shared phone held by a working parent:** record each family's preferred call time, schedule evening calls and keep voice lessons available on demand.
- **Teacher workload:** automatic packaging of lessons, volunteer mentors for calls and a cap of about one extra hour per week.
- **Privacy and child safeguarding:** collect only name, class and guardian number; obtain written parental consent; use role-based access and encryption; call only the guardian's phone; vet mentors; comply with Bangladeshi data protection rules.
- **Cost and power:** SMS and voice charges may grow at scale, so negotiate zero-rating and fall back to worksheets; pre-distribute printed packs before monsoon season in case of power cuts.

Limitations: ShikkhaSetu supports, but cannot replace, classroom teaching; voice and SMS suit literacy and numeracy better than practical subjects; and its impact in Bangladesh is unproven until piloted. **Next steps:** (1) interview 10 teachers and 20 parents in one upazila to test the assumptions; (2) build a clickable prototype; (3) test a voice lesson and SMS quiz with 30 students; (4) approach a2i or an education NGO as a pilot partner.

8. References and Sources

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